

Mathematics Backbone Crosswalk for Anonymous-DHT Receipts

What the five published wiki notes export, and where to cite them

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Abstract

This note is a non-redundancy aid. The One-True-Archive relies on five already-published mathematics notes (canonical wiki links) for core mechanism design and lower bounds. Downstream papers should not repeatedly re-explain that machinery. We therefore record a compact crosswalk: what each backbone note exports as an interface, which series papers depend on it, and how to chain the exports into receipt-grade line items and endpoint sizing. This is not a survey and does not introduce new primitives.

Scope. A citable index for the five published mathematics notes (wiki links) and how they connect to the archive’s receipt and certification interfaces.

Imports. Receipt vocabulary is imported from Synthesis 12; equivalence/drift rules from Synthesis 18; endpoint sizing from Odds Inflation (Anonymity A) and the Endpoint Bridge (Synthesis 5). For delegation bounds and certification hooks we rely on Spectral Delegation Certificates (Anonymity D) and Regime II conditional certificates (Synthesis 21). [1, 2, 7, 8, 9, 10]

Exports. Exports (i) a “what-to-cite-where” table for the backbone notes and (ii) a minimal budget pipeline sketch that avoids re-derivations in downstream papers.

Canonical wiki links (published backbone).

- [[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]
- [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]
- [[2026.01.25 - Mathematics: Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding]]
- [[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]]
- [[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]]

Recommended citation keys. To keep downstream L^AT_EX papers uniform, we recommend citing the backbone notes via the following stable bibkeys (already provided in this note’s bibliography):

Bibkey	Wiki link (canonical)
wiki:scheduling	[[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]
wiki:spectral	[[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]
wiki:prefixcap	[[2026.01.25 - Mathematics: Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding]]
wiki:stopadd	[[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]]
wiki:knapsack	[[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]]

Downstream papers should cite this crosswalk for *where* the exports land in receipts, and cite the relevant `wiki:*` entry for the mechanism/lower-bound content.

1 What each backbone note exports (as an interface)

We treat each backbone note as exporting a small set of reusable objects. Downstream papers should cite the relevant backbone note for the *mechanism/lower-bound* content and cite the synthesis layer for the *receipt surface*.

1.1 2026-01-22: Scheduling/compiler techniques

Exports. (i) public-schedule and stop-time-padding models; (ii) compiler-oriented constraints (calendar/poset feasibility); (iii) TV-style privacy certificates for termination-time / deadline-hiding mechanisms. **Receipt-facing touchpoints.** threat-window declarations (Synthesis 4), termination-time/stop-time exposure normal forms (Synthesis 3), and TV/dual-certificate accounting (published 2026-01-29; Congestion-EQ).[4, 3, 18]

1.2 2026-01-23: Spectral anonymity for delegation

Exports. Random-walk delegation distinguishing bounds expressed via spectral witnesses (including directed overlays), and a clean “per-step leakage” packaging that can be attached to a receipt as evidence. **Receipt-facing touchpoints.** Anonymity D packages the spectral witness as a receipt-grade MaxL/odds-inflation budget; Synthesis 21 provides the conditional-certificate shape when the bound is assumption-qualified.[9, 10]

1.3 2026-01-25 and 2026-01-26: Prefix capacities and stop-time normal forms

Exports. A termination-time hiding frontier under shared-kernel constraints: prefix-capacity bounds, separation cuts, and normal forms for deadline distributions. **Receipt-facing touchpoints.** These results justify which termination-time witness families are actually equalizable under an operational kernel constraint; the exposure IDs for those families live in Synthesis 3 and the repetition/window handling lives in Synthesis 4.[3, 4]

If you need ...	cite ...
A stop-time / termination-time padding mechanism and its TV guarantee	Scheduling/compiler (2026-01-22) + (when using normal forms) Stop-time addendum (2026-01-26) + receipt-facing trace/TW definitions (Synthesis 3–4).
A feasible padding optimization with published certificates	Poset-feasible padding + duals (2026-01-29) + proof-carrying budgets (Boss Fight C).
A random-walk delegation anonymity bound for an overlay	Spectral anonymity (2026-01-23) + spectral certificates (Anonymity D) + endpoint sizing (Synthesis 5).
When the bound is assumption-qualified (e.g., separation, state slice)	Conditional certificates (Synthesis 21) + separation manifests (Synthesis 22) or state declarations (Synthesis 18 / State series).
A quick pointer for operators/readers instead of re-deriving math	Cite this crosswalk (Synthesis 24) + the specific backbone note.

Table 1: Crosswalk from backbone mathematics notes to receipt-facing interfaces in this archive.

1.4 2026-01-29: Optimal poset-feasible padding (knapsack/transport + dual certificates)

Exports. A TV-privacy optimization stack for padding under feasibility constraints (knapsack/transport) together with dual certificates that can be published as replayable evidence.

Receipt-facing touchpoints. Use this note when you want a termination-time hiding claim to be accompanied by a dual certificate (proof-carrying budget style) rather than only a point estimate.[18, 11]

2 What to cite where (crosswalk)

Table 1 is the main non-redundancy deliverable.

3 Minimal pipeline sketch (how the pieces compose)

This section is intentionally short: it is here so downstream papers can cite a single paragraph rather than re-stitching the same chain.

Delegation slice (spectral $\Rightarrow$ endpoint). A spectral witness bound from 2026-01-23 yields a per-step leakage bound (Anonymity D) that can be published as a receipt line item bound b in bits (MaxL or odds inflation), possibly conditional on a state/separation guard. For a horizon n (or a declared retry window), compose using the checkpoint accountant (Anonymity A / Synthesis 19) and translate to endpoint sizing using the bridge rule (Synthesis 5).[9, 7, 6, 8]

Stop-time padding slice (TV $\Rightarrow$ receipt-grade evidence). A stop-time padding mechanism (2026-01-22/26) yields a TV-style guarantee on a declared termination-time witness family. If the deployment uses feasibility constraints (calendar/poset), 2026-01-29 provides an optimization and a dual certificate that can be committed via ABOM/OINL and replayed (proof-carrying budgets). Receipt-facing objects are: the exposure normal-form id (Synthesis 3), the threat-window id (Synthesis 4), and the replay hook binding the certificate plan spec (Synthesis 12/18/20).[3, 4, 1, 2, 5, 14, 17, 18]

4 Research prompts (where more work seems warranted)

The backbone notes provide strong local results; several cross-paper “glue” problems remain worthwhile:

- **End-to-end composition across heterogeneous witnesses.** Provide a single receipt-grade theorem that composes a delegation bound (spectral) with a stop-time padding bound (TV) under one threat-window declaration.
- **Directed/weighted overlays in deployment.** The spectral note treats directed overlays; the series would benefit from a deployment-facing canonicalization of “what graph object is committed” when the overlay is dynamic.
- **Certificate compression and witness minimization.** Dual certificates can be large; connecting the routing-signature compression work to padding/TV certificates is a practical next step.[12]
- **Separation/failure modeling.** For separation-qualified primary paths, publishing a credible ρ remains operationally hard; the probabilistic separation note (Synthesis 23) lists evidence shapes, but calibrating them against modern reconnaissance and compromise rates is an open deployment discipline problem.[13]

References

- [1] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Series synthesis note (Synthesis 12), 2026. (This archive.)
- [2] Anonymous. Plan and State Equivalence: Digest IDs, Drift Classes, and Replay Semantics. Series synthesis note (Synthesis 18), 2026. (This archive.)
- [3] Anonymous. Trace Interfaces for Anonymous DHTs: Observation Tiers, Committee-Contact Traces, and Exposure Normal Forms. Series synthesis note (Synthesis 3), 2026. (This archive.)
- [4] Anonymous. Threat Windows and Declared Resets for Anonymous DHT Budgets. Series synthesis note (Synthesis 4), 2026. (This archive.)
- [5] Anonymous. Auditor Replay Recipe: A Mechanical Checklist for Reproducing Anonymity Receipts. Series synthesis note (Synthesis 20), 2026. (This archive.)
- [6] Anonymous. Receipt Accounting Rulebook: Checkpoints, Composition, and Ledger-Friendly Summaries. Series synthesis note (Synthesis 19), 2026. (This archive.)
- [7] Anonymous. Odds Inflation, Privacy Loss, and Endpoint Anonymity Bounds. Anonymity series Paper A, 2026. (This archive.)

- [8] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Series synthesis note (Synthesis 5), 2026. (This archive.)
- [9] Anonymous. Spectral Delegation for Anonymous DHT Lookups: From mixing certificates to receipt-grade per-step budgets. Anonymity series Paper D, 2026. (This archive.)
- [10] Anonymous. Conditional Per-Step Budget Certificates (Regime II): Assumption-Qualified Bounds in Anonymous DHTs. Series synthesis note (Synthesis 21), 2026. (This archive.)
- [11] Anonymous. Proof-Carrying Budgets for Anonymous DHT Lookups. Boss Fight series Paper C, 2026. (This archive.)
- [12] Anonymous. Routing Signature Compression for Certified DHT Budgets. Certified series Paper B, 2026. (This archive.)
- [13] Anonymous. Probabilistic Separation Budgets: Separation Failure as a Receipt-Grade $(0, \rho)$ Adjunct. Series synthesis note (Synthesis 23), 2026. (This archive.)
- [14] Anonymous. Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups. Published mathematics note (2026-01-22). Wiki: `[[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]`.
- [15] Anonymous. Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays. Published mathematics note (2026-01-23). Wiki: `[[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]`.
- [16] Anonymous. Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding. Published mathematics note (2026-01-25). Wiki: `[[2026.01.25 - Mathematics: Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding]]`.
- [17] Anonymous. Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms. Published mathematics note (2026-01-26). Wiki: `[[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]]`.
- [18] Anonymous. Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy. Published mathematics note (2026-01-29). Wiki: `[[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]]`.